

Turing Machines In Class Examples and Extra Problems

1. Design a Turing machine that recognizes the following languages.
 - (a) A language that can be represented using the regular expression $r = 0^+$.
 - (b) A language that can be represented using the regular expression $r = a(a \cup b)^*$.
 - (c) A language that can be represented using the regular expression $r = aba^*b$.
 - (d) $L = \{w : w \in \{a, b\}^*, |w| \text{ is odd}\}$
 - (e) $L = \{a^n : n \neq 2 \text{ and } n \neq 4\}$
 - (f) $L = \{a^n b^n : n > 0\}$
 - (g) $L = \{w \in \{a, b\}^* : n_a(w) \neq n_b(w)\}$
 - (h) $L = \{a^n b^m : n < m\}$
 - (i) $L = \{a^n b^m a^{n+m} : n \geq 0, m \geq 1\}$
2. Design a Turing Machine that transforms a string of a 's and b 's to a string of b 's only.
3. Design a Turing Machine with a tape full of a 's and b 's, that appends an a at the end.
4. Design a Turing Machine with a tape full of a 's and b 's, that replaces the 1st a with a b . Please make sure that the read head ends at the end of the input string on the tape.
5. Design a Turing Machine with a tape full of a 's and b 's, that replaces all a 's except for the leftmost one on the tape with b 's.
6. Design a Turing Machine with a tape full of a 's and b 's, that replaces two consecutive b 's with a 's.
7. Design a Turing Machine with a tape full of a 's and b 's, that **erases** the input completely.
8. Design a Turing Machine with a tape full of a 's and b 's, that **erases** the b 's only. For example, if the tape input contained $abbaaba$, then the tape should contain $aaaa$.
9. Design a Turing Machine that copies a string of 1's: $q_0 w \vdash^* q_f w w$ for any $w \in \{1\}^+$.

10. Let x and y represent positive integers in unary notation. Design a Turing Machine that computes $x + y$.

Example:

$$x = 3 \quad w(x) = 111$$

$$y = 2 \quad w(y) = 11$$

Input $w(x)0w(y)$

Output $w(x + y)0$

Computation

$$q_0w(x)0w(y) \vdash^* q_fw(x + y)0$$

11. Let x and y represent positive integers in unary notation. Design a Turing Machine that computes $x - y$ with the assumption that $x > y$.

Example:

$$x = 5 \quad w(x) = 11111$$

$$y = 2 \quad w(y) = 11$$

Input $w(x)0w(y)$

Output $w(x - y)$

Computation

$$q_0w(x)0w(y) \vdash^* w(x - y)q_f$$

12. Given a and b that represent two positive integers in unary notation. Design a Turing Machine that will halt in a final state if $a \geq b$ and a non-final state if $a < b$.

Computation

$$q_0w(a)0w(b) \vdash^* q_yw(a)0w(b) \text{ if } a \geq b$$

$$q_0w(a)0w(b) \vdash^* q_nw(a)0w(b) \text{ if } a < b$$

13. Give an implementation-level description of the following Turing Machine that decides the following language over the alphabet $\Sigma = \{0, 1\}$.

$$L = \{w \mid w \text{ contains an equal number of 0s and 1s}\}$$

Note: You are not constructing a transition graph. Please only provide the implementation-level description.